Ziheng Xiao

Date of Birth : Mar. 9, 1995
Area of Specialism : Power Electronics, Artificial Intelligence.
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Education and Working Experience:

Year	Degree	Discipline	Institution
2017-2022	Doctor of Philosophy (supervised by Professor *An Luo*, member of Chinese engineering academy)	Electrical Engineering	Hunan University
2013-2017	Bachelor	Electrical Engineering and Automation	Hunan University
Year	**Working Experience**		
2023-Present	Part-time Lecturer of School of Electrical and Electronic Engineering (EEE), NTU EE6506 Semiconductor Based Converter in Renewable Energy Systems EE6514 Special Topics in Clean Energy System Design EE6501 Power Electronic Converters		
2022-Present	Research Fellow (supervised by Associate Professor *Yi Tang*, Cluster Director of ERI@N, Associate Chair (Graduate Studies), EEE, NTU)		

Project Participated:

Year	Projects	Direct costs
2026-2030	Efficiency Improvement Advanced Flow-based Energy Storage Innovations for Scalability & Systemic Industry Collaboration Projects (IAF-ICP), Singapore	S$ 16.5 M
2024-2027	Advancing Electrified Transportation and Intelligent Systems through Physics-Informed Machine Learning in Wireless Power Transfer (R24I6IR134), Agency for Science, Technology and Research (A*STAR), Singapore	S$ 350 k
2024-2027	Optimizing Wireless Power Transfer Efficiency Through Transfer Learning-Based Precise Power Loss Modelling (RG73/24), Ministry of Education (MOE), Singapore	S$ 160 k
2021-2025	Advanced Energy Management and Data Fusion for Smart Homes and Smart Grids, Lite-On Singapore Pte. Ltd., Singapore	S$ 5 M
2019-2021	Natural Science Foundation of China under Grant (51807057), China	¥ 240 k
2019-2021	Hunan Province Youth Fund Project (2019JJ50038), China	¥ 50 k
2018-2023	Project of Megmeet (University-Industry Collaboration Programme), China	¥ 1.2 M
2018-2020	Hunan Science and Technology Innovation Plan Project (2017XK2104), China	¥ 5 M

Citation Summary:

Database	Citation Count		H-index
	without self-citations	with self-citations	
Scopus		430	13
Web of Science (SCI)	200	261	10
Google Scholar		455	13

Publication Summary:

Journal	Publication Count					Total 37
	TIE	TPEL	TTE	JESTPE	Others	
First Author	3	14	1	2	0	20
Corresponding Author	0	2	1	0	0	3
Co-Author	3	7	1	2	1	14
Conference	**Publication Count**					**Total 27**
	APEC	**ECCE**	**IECON**	**ECCE-Asia**	**Others**	
First Author	0	2	2	2	3	9
Co-Author	5	3	1	4	5	18

Journal Papers, First Author:

	Publication Title	IF: 135.1
1	**Z. Xiao**, Z. He, R. Guan and A. Luo, "Piecewise-Approximated Time Domain Analysis of LLC Resonant Converter Considering Parasitic Capacitors and Deadtime," in IEEE Transactions on Power Electronics, vol. 38, no. 1, pp. 578-592, Jan. 2023, doi: 10.1109/TPEL.2022.3207329.	6.7
2	**Z. Xiao**, Y. Jiang, T. Sun, Y. Wu and Y. Tang, "Refining Power Converter Loss Evaluation: A Transfer Learning Approach," in IEEE Transactions on Power Electronics, vol. 39, no. 4, pp. 4313-4324, April 2024, doi: 10.1109/TPEL.2023.3349178.	6.7
3	**Z. Xiao**, Z. He, H. Wang, A. Luo, Z. Shuai and J. M. Guerrero, "General High-Frequency-Link Analysis and Application of Dual Active Bridge Converters," in IEEE Transactions on Power Electronics, vol. 35, no. 8, pp. 8673-8688, Aug. 2020, doi: 10.1109/TPEL.2019.2963746.	6.7
4	**Z. Xiao**, Z. He, Z. Li, L. Zhu and L. Wang, "Unified Description and Optimization Method of Dual Active Bridge DC–DC Converters," in IEEE Transactions on Power Electronics, vol. 37, no. 10, pp. 11839-11854, Oct. 2022, doi: 10.1109/TPEL.2022.3177245.	6.7
5	**Z. Xiao**, Z. Li, Z. He, H. Wang, Y. Tang and A. Luo, "Computer-Aided Time-Domain Operation and Design Optimization of Resonant Converters," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 11, no. 4, pp. 4164-4177, Aug. 2023, doi: 10.1109/JESTPE.2023.3275637.	5.5
6	**Z. Xiao**, Z. He, R. Guan, Z. Li, B. Zhou and A. Luo, "A Three-Terminal Submodule Based High DC Conversion Ratio System With Self-Balance Feature," in IEEE Transactions on Power Electronics, vol. 37, no. 5, pp. 5650-5663, May 2022, doi: 10.1109/TPEL.2021.3131795.	6.7
7	**Z. Xiao**, Y. Jiang, F. Deng, Z. Yao and Y. Tang, "A Data-Driven Control Parameters Optimization Method for Dual Active Bridge Converters," in IEEE Transactions on Industrial Electronics, vol. 71, no. 11, pp. 14054-14066, Nov. 2024, doi: 10.1109/TIE.2024.3370950.	7.7
8	**Z. Xiao** et al., "A Hybrid Data-Driven Power Loss Minimization Method of Dual-Active Bridge Converters," in IEEE Transactions on Power Electronics, vol. 39, no. 5, pp. 5820-5832, May 2024, doi: 10.1109/TPEL.2024.3368330.	6.7
9	**Z. Xiao**, Z.Yu, and Y. Tang, " Swift and Seamless Start-Up of DAB Converters in Constant and Variable Frequency Modes," in IEEE Transactions on Industrial Electronics, vol. 72, no. 5, pp. 4763-4775, May 2025, doi: 10.1109/TIE.2024.3481900.	7.7
10	**Z. Xiao** et al., "Analysis and Design of a Modular Self-Balance High-Voltage Input DC–DC Topology," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 10, no. 2, pp. 2276-2289, April 2022, doi: 10.1109/JESTPE.2022.3140898.	5.5

11	**Z. Xiao**, X. Li and Y. Tang, "A Lightweight Artificial Neural Network Start-Up Controller for CLLC Resonant Converters," in IEEE Transactions on Power Electronics, vol. 39, no. 11, pp. 14775-14786, Nov. 2024, doi: 10.1109/TPEL.2024.3436847.	6.7
12	**Z. Xiao**, Z. Yao, F. Deng, L. Zhang and Y. Tang, "Seamless Rotational Phase Shedding for Multiphase DC–DC Converters," in IEEE Transactions on Power Electronics, vol. 39, no. 5, pp. 4996-5001, May 2024, doi: 10.1109/TPEL.2024.3359604.	6.7
13	**Z. Xiao**, M. Xiao, Z. He, L. Wang, Z. Li and H. Wang, " Light Load Efficiency Improvement of DAB Converters With Optimal Switching Sequences," in IEEE Transactions on Power Electronics, vol. 40, no. 7, pp. 9343-9356, July 2025, doi: 10.1109/TPEL.2024.3513412.	6.7
14	**Z. Xiao** et al., "Exploring Optimal Switching Patterns for Rapid Start-Up in Synchronous Boost Converters," in IEEE Transactions on Industrial Electronics, vol. 72, no. 8, pp. 7897-7909, Aug. 2025, doi: 10.1109/TIE.2024.3522511.	7.7
15	**Z. Xiao**, Z. Yao, F. Deng, Y. Zhang and Y. Tang, "Exploring the Extreme Dynamic Transition Response of Dual Active Bridge Converters," in IEEE Transactions on Power Electronics, vol. 40, no. 5, pp. 6988-7001, May 2025, doi: 10.1109/TPEL.2025.3532400.	6.7
16	**Z. Xiao**, F. Deng, Z. Yao and Y. Tang, " Soft-Morphing: Unlocking the Potential of CLLC Converters for Ultrawide Range Applications," in IEEE Transactions on Power Electronics, vol. 40, no. 1, pp. 1289-1304, Jan. 2025, doi: 10.1109/TPEL.2024.3480909.	6.7
17	**Z. Xiao,** J. Wu, Z. He, Z. Yao and Y. Tang, "Optimal Switching Pattern Control of the CLLC Resonant Converters in Dynamic Processes," in IEEE Transactions on Power Electronics, vol. 39, no. 12, pp. 16268-16282, Dec. 2024, doi: 10.1109/TPEL.2024.3457899.	6.7
18	**Z. Xiao**, F. Deng, Z. Yao and Y. Tang, "Minimizing and Balancing Power Losses in Multiphase Paralleled Bridge Legs," in IEEE Transactions on Power Electronics, vol. 40, no. 9, pp. 13835-13851, Sept. 2025, doi: 10.1109/TPEL.2025.3569667.	6.7
19	**Z. Xiao**; Z. He; L. Zhang; Z, Yao, and Y. Tang, " "Resonant DC Transformer with Integer Multiple Frequency for High Step-Ratio Application," in IEEE Transactions on Transportation Electrification, doi: 10.1109/TTE.2025.3627551.	7.2
20	**Z. Xiao**, T. Sun, Z. Yao and Y. Tang, "Optimizing Multi-phase DC-DC Converters via Inter-leaving/Intra-leaving and Phase Shedding," in IEEE Transactions on Power Electronics, doi: 10.1109/TPEL.2025.3631793.	6.7

Journal Papers, Corresponding Author:

	Publication Title	IF: 20.6
1	Y. Zeng; **Z. Xiao**; Q. Liu; G. Liang; E. Rodriguez; G. Zou; X. Zhang; J. Pou " "Physics-Informed Deep Transfer Reinforcement Learning Method for the Input-Series Output-Parallel Dual Active Bridge-Based Auxiliary Power Modules in Electrical Aircraft," in IEEE Transactions on Transportation Electrification, vol. 11, no. 2, pp. 6629-6639, April 2025, doi: 10.1109/TTE.2024.3514657.	7.2 **ESI Highly Cited Paper**
2	Z. Yao; X. He; **Z. Xiao**; F. Deng; C. Dai; S. Lu; Y. Tang " Current Ripple Prediction and ZVS-Based Variable Switching Frequency Control for Interleaved Multiphase Three-Level DC–DC Converter," in IEEE Transactions on Power Electronics, vol. 40, no. 4, pp. 5109-5119, April 2025, doi: 10.1109/TPEL.2024.3521997.	6.7
3	Z. Yao, L. Jiang, T. Sun, **Z. Xiao**, W. Chen and Y. Tang, " Smooth Optimal Trajectory and Voltage Balancing Control for T-Type Three-Level LLC Resonant Converter in Alternating Mode," in IEEE Transactions on Power Electronics, vol. 40, no. 7, pp. 9278-9288, July 2025, doi: 10.1109/TPEL.2025.3548174.	6.7

Journal Papers, Co-Author:

	Publication Title	IF: 93.2
1	Z. Yao, **Z. Xiao** et al., "Sensorless Simultaneous Self-Balance Mechanism of Voltage and Current Based on Near-CRM in Interleaved Three-Level DC–DC Converter," in IEEE Transactions on Power Electronics, vol. 39, no. 2, pp. 2086-2099, Feb. 2024, doi: 10.1109/TPEL.2023.3330812.	6.7
2	M. Chen et al., "MagNet Challenge for Data-Driven Power Magnetics Modeling," in IEEE Open Journal of Power Electronics, vol. 6, pp. 883-898, 2025, doi: 10.1109/OJPEL.2024.3469916, doi: 10.1109/OJPEL.2024.3469916.	5.0
3	Z. Yao, X. He, H. Lan, H. Yang, **Z. Xiao** et al., "A Trapezoidal Current Mode to Reduce Peak Current in Near-CRM for Three-Level DC–DC Converter," in IEEE Transactions on Power Electronics, vol. 39, no. 8, pp. 9446-9456, Aug. 2024, doi: 10.1109/TPEL.2024.3399216.	6.7
4	F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, " Decentralized Multilayer Master-Slave Control Strategy for Power Management in Autonomous DC Microgrid Clusters," in IEEE Transactions on Industrial Electronics, vol. 72, no. 3, pp. 2712-2723, March 2025, doi: 10.1109/TIE.2024.3443949.	7.7
5	F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, "Inertia Emulation in Droop-Based DC Microgrids With Equivalent Converter Impedance Reshaping," in IEEE Transactions on Power Electronics, vol. 39, no. 11, pp. 15206-15216, Nov. 2024, doi: 10.1109/TPEL.2024.3437767.	6.7
6	R. Hou; Y. Liu; Z. He; Z. Li; M. Xiao; J. Guan; Y. Chen; J. Wang; **Z. Xiao**; H. Wang "Optimization and Design of LCC Resonant High-Voltage Wide Range Power Supply for Magnetrons," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 12, no. 4, pp. 3835-3847, Aug. 2024, doi: 10.1109/JESTPE.2024.3412800.	5.5
7	Y. Li, S. Wang, Y. Wu, Y. Jiang, **Z. Xiao** and Y. Tang, " Heterogeneous Integration of Isotropic and Anisotropic Magnetic Cores for Inductive Power Transfer," in IEEE Transactions on Power Electronics, vol. 40, no. 2, pp. 3770-3784, Feb. 2025, doi: 10.1109/TPEL.2024.3485193.	6.7
8	Z. Yao, X. He, M. Liu, J. Liu, **Z. Xiao** and Y. Tang, " Fixed Switching Frequency Control Using Trapezoidal Current Mode to Achieve ZVS in Three-Level DC–DC Converters," in IEEE Transactions on Industrial Electronics, vol. 72, no. 4, pp. 4175-4185, April 2025, doi: 10.1109/TIE.2024.3454707.	7.7
9	H. Yang, F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, "Power System Inertia Enhancement Based on LVDC Microgrids," in IEEE Transactions on Industrial Electronics, doi: 10.1109/TIE.2024.3508069.	7.7
10	Y. Wu; Y. Jiang; Y. Li; **Z. Xiao**; H. Yuan; X. Wang; Y. Tang " Precise Coil Inductance Prediction in WPT Systems: A Transfer Learning Approach," in IEEE Transactions on Transportation Electrification, vol. 11, no. 2, pp. 7083-7095, April 2025, doi: 10.1109/TTE.2024.3524106.	7.2
11	T. Sun, **Z. Xiao**, F. Deng, Z. He, X. Pan and Y. Tang, "Single-Stage Multiport AC–DC–DC Converter With Reactive Power Control Capability," in IEEE Transactions on Power Electronics, vol. 40, no. 5, pp. 6700-6713, May 2025, doi: 10.1109/TPEL.2025.3532439.	6.7
12	Z. Yao, **Z. Xiao**, et al., "Generalized Trapezoidal Current Mode-Based Zero-Voltage Switching for Multilevel DC–DC Converters," in IEEE Transactions on Power Electronics, vol. 40, no. 7, pp. 8956-8961, July 2025, doi: 10.1109/TPEL.2025.3551341.	6.7
13	G. Luo, Z. Yao, L. Jiang, **Z. Xiao**, W. Chen and Y. Tang, "Current Balancing Control Method With Reduced Number of Current Sensors for Interleaved LLC Converters," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 12, no. 3, pp. 2604-2613, June 2024, doi: 10.1109/JESTPE.2024.3388481.	5.5
14	X. He, Z. Yao; X. Yang; B. Ngo; **Z. Xiao** et al., "Hybrid TZCM–QCM Control Method for Smooth Power Transition and ZVS in a Bidirectional Three-Level DC-DC Converter," in IEEE Transactions on Power Electronics, doi: 10.1109/TPEL.2025.3630091.	6.7

Conference Papers, First Author:

	Publication Title	P.s.
1	**Z. Xiao**, Z. Yao, Y. Zeng and Y. Tang, "Light Load Audible-Noise-Free Skip Control for Dual Active Bridge Converters," 2025 IEEE 15th International Conference on Power Electronics and Drive Systems (PEDS), Penang, Malaysia, 2025, pp. 1-5, doi: 10.1109/PEDS63958.2025.11144783.	
2	**Z. Xiao**, L. Zhang, F. Deng and Y. Tang, "Comparisons of Cycle-by-Cycle Start-Up Control in Synchronous Boost Converters," 2025 IEEE 15th International Conference on Power Electronics and Drive Systems (PEDS), Penang, Malaysia, 2025, pp. 1-5, doi: 10.1109/PEDS63958.2025.11144889.	
3	**Z. Xiao**, Y. Jiang, C. Wang and Y. Tang, "A Streamlined and Intuitive AI-Powered Analysis and Design Tool for Resonant Converters," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 852-857.	Best Paper Award First Prize **3/927**
4	**Z. Xiao**, Y. Jiang, Z. Yao and Y. Tang, "AI-assisted Peak Efficiency Searching of CLLC Resonant Converters in Step-down Conditions," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 863-868.	
5	**Z. Xiao**, L. Zhang, Z. He and Y. Tang, "Efficiency Improvement of DAB Converters with Enhanced Inductor Current Classification," 2023 IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, TN, USA, 2023, pp. 2631-2638.	
6	**Z. Xiao**, Z. He, F. Deng and Y. Tang, "Power Loss Minimization of CLLC Resonant Converters via Time Domain Analysis," 2023 IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, TN, USA, 2023, pp. 3392-3399.	
7	**Z. Xiao**, Y. Jiang, Y. Jiang and Y. Tang, "A Zero-Knowledge ANN-Based Waveform and Critical Parameter Calculator for Resonant Converters," IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society, Singapore, Singapore, 2023, pp. 1-6.	
8	**Z. Xiao**, Y. Jiang, Z. Yao and Y. Tang, "A Precise and Fast BPNN-Based Voltage Gain Model of CLLC Converters in All Operation Conditions," IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society, Singapore, Singapore, 2023, pp. 1-6.	
9	**Z. Xiao**, Z. He, A. Luo, Q. Xu, X. Huang and J. Zhang, "A Modular High-Frequency Resonant DC-DC Transformer for MVDC Application," 2018 IEEE International Power Electronics and Application Conference and Exposition (PEAC), Shenzhen, China, 2018, pp. 1-6.	

Conference Papers, Co-Author:

	Publication Title	P.s.
1	D. Zhao, Y. Wu, Y. Li, R. Zhang, **Z. Xiao** and Y. Tang, "A Convenient Harmonic State Space Model for ZVS Analysis in Wireless Power Transfer Systems," 2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025, pp. 1-6, doi: 10.1109/ECCE58356.2025.11259601.	
2	Y. Zeng, J. Pou; G. Zou; H. Jie; **Z. Xiao**, et al., "Physics-Informed Neural Network for DC Solid State Transformers in DC Microgrids," IECON 2025 – 51st Annual Conference of the IEEE Industrial Electronics Society, Madrid, Spain, 2025, pp. 1-6, doi: 10.1109/IECON58223.2025.11221331.	
3	Z. Yao, J. Liu, S. Yao, B. Ngo, **Z. Xiao** and Y. Tang, " Exploring the Soft-Switching Benefits of TZCM Mode in Three-Level DC-DC Converters Using Wide Bandgap Power Devices," 2025 IEEE Workshop on Wide Bandgap Power Devices and Applications in Asia (WiPDA Asia), Beijing, China, 2025, pp. 1-5, doi: 10.1109/WiPDA-Asia63772.2025.11183701.	
4	Z. Yao, J. Liu, F. Deng, **Z. Xiao**, Y. Jiang and Y. Tang, "A Five-Level Buck Converter with Zero-Voltage Switching Based on Trapezoidal Current Mode," 2025 IEEE 15th International Conference on Power Electronics and Drive Systems (PEDS), Penang, Malaysia, 2025, pp. 1-5, doi: 10.1109/PEDS63958.2025.11144861.	
5	Z. Yao, X. He, **Z. Xiao**, F. Deng, W. Chen and Y. Tang, "A Novel Trapezoidal Current Mode for Realizing Zero-Voltage Switching in Three-Level DC-DC Converters," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 4894-4899.	
6	F. Deng, S. Zhang, L. Zhang, Z. Yao, **Z. Xiao** and Y. Tang, "Decentralized Current Sharing in Islanded DC Microgrids Based on AC Frequency Droop," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 1364-1369.	
7	L. Zhang, **Z. Xiao**, H. Wang, Y. Tang, S. Yang and X. Qi, "Analyze and Mitigate Low-Frequency Ripple of Arm Current for Modular Multilevel Converter based Solid-State Transformers," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 1108-1113.	
8	L. Zhang, **Z. Xiao**, F. Deng, Z. Yao, H. Yang and Y. Tang, "Grid-Side High-Order Harmonics Mitigation for Three-Phase Four-Wire Inverter Based on Repetitive Control and Virtual Filter," 2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia), Chengdu, China, 2024, pp. 1079-1083.	
9	Z. Yao, X. He, **Z. Xiao**, Z. He, Y. Jiang and Y. Tang, "Full-Range ZVS Control Method Based on Near-CRM for T-Type Three-Level Inverter," 2023 IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, TN, USA, 2023, pp. 2987-2992.	
10	Y. Jiang, Y. Wu, Y. Li, Z. He, **Z. Xiao**, H. Wang;X. Wang and Y Tang "Ultra-high Efficiency Wireless Power Transfer Systems Based on Novel Multi-variable Control Strategy," 2023 IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, TN, USA, 2023, pp. 1659-1665.	
11	Y. Wu, Y. Jiang, Y. Li, Z. He;C. Wang, **Z. Xiao**, X. Wang and Y Tang, "A Universal Coil Structure Design Method with Accurate Numerical Computation in Wireless Power Transfer Systems," 2023 IEEE Energy Conversion Congress and Exposition (ECCE), Nashville, TN, USA, 2023, pp. 6497-6504.	
12	Y. Jiang, Y. Wu, Y. Li, **Z. Xiao**, N. Wang, M. Wu, X. Wang and Y. Tang, "Precise Modeling for the Inductance of Rounded Rectangular Coils in Wireless Power Transfer Systems," IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society, Singapore, Singapore, 2023, pp. 1-7.	
13	T. Sun, **Z. Xiao**, Z. He and Y. Tang, "Single-Stage Isolated Three-Port AC-DC-DC Converter with Asymmetrical Modulation," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 501-505.	
14	Z He, H Ding, Z Zhang, J Wu, D Zhang, J Shao, **Z Xiao**, Z Yao, Y Tang "A Dynamic Sampling Method to Achieve Noise-Free Sampling for High-Frequency Converters," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 2789-2793.	
15	L Zhang, H Yang, **Z Xiao**, Z Yao, F Deng, T Sun, Y Tang, "Capacitor Voltage-Balancing	

	Method of Three-Phase Four-Wire T-Type Inverter Based Active Power Filter," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 2277-2281.	
16	Y Li, S Wang, Y Wu, Y Jiang, X Zhu, **Z Xiao**, Z He, Y Tang, "A Novel Hybrid Magnetic Core Design Method for Weight Reduction of Wireless Power Transfer Systems," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 1202-1206.	
17	Z He, H Ding, Z Zhang, J Wu, D Zhang, YZ Liu, **Z Xiao**, Z Yao, Y Tang "Coordinate Control of NP Voltage Balance in Two-Stage Three-level DC-AC Converters," 2024 IEEE Applied Power Electronics Conference and Exposition (APEC), Long Beach, CA, USA, 2024, pp. 2749-2755	
18	F. Deng, H. Wang, **Z. Xiao**, L. Zhang, Z. Yao and Y. Tang, "Decentralized Master-Slave Control Strategy for Current Sharing in Islanded DC Microgrids," 2024 IEEE 9th Southern Power Electronics Conference (SPEC), Brisbane, Australia, 2024, pp. 1-5, doi: 10.1109/SPEC62217.2024.10893277.	

Patents filed:

	Publication Title	P.s.
1	A. Luo, **Z. Xiao** *et al.,* SWITCHED CAPACITOR RESONANT VOLTAGE-MULTIPLYING RECTIFIER CONVERTER, AND A CONTROL METHOD AND A CONTROL SYSTEM THEREOF. WO2023005270A1.	
2	A. Luo, **Z. Xiao** *et al.,* SWITCHED CAPACITOR RESONANT VOLTAGE-MULTIPLYING RECTIFIER CONVERTER, AND A CONTROL METHOD AND A CONTROL SYSTEM THEREOF. WO2023005270A1.	
3	Z. He, **Z. Xiao** *et al.,* Automatic voltage balance switch network, DC converter, control system and control method. ZL202011485672. 8.	
4	A. Luo, **Z. Xiao** *et al.,* Switched capacitor resonance voltage doubler rectifier converter and its control method and control system. ZL 202110852407 .7.	
5	R. Guan, A. Luo, … **Z. Xiao** *et al.,* Modular combined medium-voltage DC converter and control method based on two-stage conversion structure, L201910868861 .4.	
6	A. Luo, **Z. Xiao** *et al.,*, Resonant DC converter with series input and multi-port output and its control method, ZL201811235162 .8.	
7	Z. He, **Z. Xiao** *et al.,* Voltage equalizing capacitor regulator for input series structure and its control method, ZL202110849483 .2.	
8	Z. He, **Z. Xiao** *et al.,* A DC-DC conversion system and its control method, ZL202110842999 .4.	
9	Z. He, B.Zhou,… **Z. Xiao** *et al.,* Submarine DC converter and control method, ZL202010499215 .8.	
10	A. Luo, Z. He, … **Z. Xiao** *et al.,* A high-reliability and high-power case-based medium and high voltage DC power supply, ZL202010499215 .8.	

Awards & Honors:

	Awards & Honors Title	Time
1	🏆 Presidential Scholarship for Doctoral Student of Hunan University **5/279**	2020
2	🏆 Outstanding Papers of 2021 in "Journal of Power Supply" **3/334**	2022
3	Session chair for the 49th Annual Conference of the IEEE Industrial Electronics Society	2023
4	🏆 Best paper award (First prize) in 2024 IPEMC Conference **3/927**	2024
5	Poster Session Chair of 2024 2nd Power Electronics and Power System Conference	2024
6	Special Session Panelists for the International Decentralized Energy Access Solutions Conference	2025
7	Special Session Organizer for the 15th IEEE International Conference on Power Electronics and Drive Systems	2025
8	Technical Program Committee for 2025 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC-AP)	2025
9	Special Issue Guest Editor for *Energies* Journal "Control and Optimization of Power Converters"	2025
10	Special Issue Guest Editor for *Electronics* Journal "Power Converters and Multilevel Converters"	2025

Teaching:

	Teaching Course Title	Time
1	NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2023 Fall
2	NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2024 Fall
3	NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2025 Fall
4	NTU EEE EE6501 Power Electronic Converters	2025 Fall
5	NTU EEE EE6514 Special Topics in Clean Energy System Design	2025 Fall
6	NTU EEE EE6501 Power Electronic Converters	2026 Spring
7	NTU EEE EE6501 Power Electronic Converters	2026 Fall

Students Mentoring/Co Mentoring:

	Name	Class	Thesis Project Title
1	MOU XINGYAN	M.SC.(POWER ENG.) Batch 2025	**M.SC.** High-Efficiency High-Frequency Power Converter Design for Next-Generation AI Processors
2	DUAN YILIN	M.SC.(POWER ENG.) Batch 2024	**EE6008 Collaborative Research and Development Project AY25S1-6-PE** High-Efficiency Power Converter Using Wide Bandgap Devices
3	MA QINGHAN	M.SC.(POWER ENG.) Batch 2024	
4	CHEN KA YI	M.SC.(POWER ENG.) Batch 2025	
5	LIU SHIQI	M.SC.(POWER ENG.) Batch 2025	
6	ZHOU BEICHEN	M.SC.(POWER ENG.) Batch 2024	
7	ZHAO WENPU	M.SC.(POWER ENG.) Batch 2025	**M.SC.** AI-Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
8	CHEN JUNYU	EEE Batch 2024	**URECA EEE25040** Optimizing Power Converter Design and Efficiency Through Transfer Learning
9	HE SONGRUI	EEE Batch 2024	**URECA EEE25041** Ultra-Efficient and Ultra-Compact Power Supply for AI Chips
10	EDWIN TAN YONG HAO	EEE Batch 2023	**FYP** Gate Driver Circuit Design for Wide Band Gap Power Devices
11	HU HAORAN	NG Batch 2024	**FYP** High-Efficiency and High-Density Power Converter Design

			for AI Chips in Data Centres
12	ZHANG HUAYU	NG Batch 2024	**FYP** Gate Driver Circuit Design for Wide Band Gap Power Devices
13	SIDHANTH NODU SANTHOSH	EEE Batch 2021	**FYP** Gate Driver Circuit Design for Wide Band Gap Power Devices
14	IAN QUEK YONG EN	EEE Batch 2021	**FYP** High-Efficiency and High-Density Power Converter Design for AI Chips in Data Centres
15	PANG YI HONG	EEE Batch 2023	**URECA EEE24041** Optimizing Power Converter Design and Efficiency Through Transfer Learning
16	CHERYL LYNN KOSASIH	EEE Batch 2021	**FYP A1102-241** AI-Assisted Design and Control of Next-Generation Power Electronic Converter for Enhanced Energy Efficiency and Power Density in Renewable Energy Systems
17	CUI QILONG	M.SC.(POWER ENG.) Batch 2024	**M.SC.** Soft Transition of CLLC Resonant Converters for Ultra-Wide Range Battery Charging Applications
18	ZHANG YUNPENG	M.SC.(POWER ENG.) Batch 2024	**M.SC.** Gate Driver Circuit Design for Wide Band Gap Power Devices
19	NGUYEN THANH TUNG	M.SC.(POWER ENG.) Batch 2024	**EE6008 Collaborative Research and Development Project AY24S1-50** Machine Learning - Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
20	YUNCHENG	M.SC.(POWER ENG.) Batch 2024	
21	WEN YUHENG	M.SC.(POWER ENG.) Batch 2024	
22	DENG MIAO	M.SC.(POWER ENG.) Batch 2024	
23	YEO YEE JIE	EEE Batch 2021	**FYP A1115-231** Inductor Design for Buck-Boost Converter
24	HUANG YIYUAN	M.SC.(POWER ENG.) Batch 2023	**M.SC.** Design and Control for a Bidirectional Interlinking Converter
25	XU DONGCHEN	M.SC.(POWER ENG.) Batch 2023	**M.SC.** Optimal Inductor Design for a 6.6kW Buck-Boost Converter
26	LU ZHIYAO	M.SC.(POWER ENG.) Batch 2023	**M.SC.** 20 kW SiC Bidirectional DC-DC Converter Design
27	LEI YIHAN	M.SC.(POWER ENG.) Batch 2022	**M.SC.** Dynamic Response Improvement of a Bidirectional Interleaved Buck-Boost Converter
28	JIN SHUHAN	M.SC.(POWER ENG.) Batch 2022	**M.SC.** Efficiency Optimization of an Interleaved Buck-Boost Converter with Wide Operation Range